

Alexandroff Spaces

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1 Introduction

Topology is a foundational area of mathematics that studies properties of space that are preserved under continuous transformations. Among the various types of topologies, **Alexandroff spaces** form an important class due to their closure properties under arbitrary intersections of open sets. This report explores the fundamental definitions, characterizations, and properties of Alexandroff spaces, as well as their connections to order theory and continuity.

2 Definitions and Examples

Definition 1. Let $(X, \mathcal{A})$ be a topological space. The topology $\mathcal{A}$ is called an Alexandroff topology if arbitrary intersections of open sets are open. That is, for any family $\{O_i\}_{i \in I} \subseteq \mathcal{A}$, we have:

$$\bigcap_{i \in I} O_i \in \mathcal{A}$$

Alternatively, this can be stated dually as: any union of closed sets is closed.

Examples:

- The Sierpiński space $\{0, 1\}$ with topology $\{\emptyset, \{0\}, \{0, 1\}\}$
- Any discrete topological space
- Colpen topology where every open set is also closed

3 Minimal Open Neighborhoods

Definition 2. In a topological space (X, T) , a point $x \in X$ has a minimal open neighborhood $V(x)$ if:

$$V(x) = \bigcap_{\substack{O \in T \\ x \in O}} O$$

Theorem 1. A space (X, T) is an Alexandroff space if and only if every point in X has a minimal open neighborhood.

Proof. ($\Rightarrow$) Suppose (X, T) is Alexandroff. Then by definition, arbitrary intersections of open sets are open. For any point $x \in X$, consider the collection $\mathcal{O}_x = \{O \in T \mid x \in O\}$. Then the intersection $V(x) = \bigcap_{O \in \mathcal{O}_x} O$ is open because T is closed under arbitrary intersections.

($\Leftarrow$) Suppose every $x \in X$ has a minimal open neighborhood $V(x)$. Let $\{U_i\}_{i \in I} \subseteq T$. Let $U = \bigcap_{i \in I} U_i$. For any $x \in U$, since each $x \in U_i$, $x \in V(x) \subseteq U_i$ for all i , so $V(x) \subseteq U$. Therefore, $U = \bigcup_{x \in U} V(x)$, a union of open sets, hence open. $\square$

Proposition 1. Let (X, T) be a T_1 space. Then the following are equivalent:

1. X is an Alexandroff space
2. T is the discrete topology

Proof. Assume X is a T_1 Alexandroff space. In a T_1 space, singleton sets $\{x\}$ are closed for all $x \in X$, so their complements are open. Consider:

$$\bigcap_{y \neq x} (X \setminus \{y\}) = \{x\}$$

Since each $X \setminus \{y\}$ is open, their intersection is open by the Alexandroff property. Thus, every singleton is open, and so every subset is open, i.e., the topology is discrete. $\square$

4 Preorders Induced by a Topology

Definition 3. A binary relation $\leq$ on X is:

- A **preorder** if it is reflexive and transitive
- A **partial order** if it is a preorder and antisymmetric

Proposition 2. Let (X, T) be a topological space. Define $x \leq y \iff x \in \overline{\{y\}}$. Then $\leq$ is a preorder called the specialization preorder.

Proof. **Reflexivity:** $x \in \overline{\{x\}}$ because every point lies in the closure of itself.

Transitivity: If $x \in \overline{\{y\}}$ and $y \in \overline{\{z\}}$, then every open set containing x also intersects $\{z\}$, hence $x \in \overline{\{z\}}$. $\square$

Proposition 3. If X is an Alexandroff space, then it is T_0 if and only if $\leq$ is a partial order.

Proof. $(\Rightarrow)$ Suppose $x \leq y$ and $y \leq x$. Then $\overline{\{x\}} = \overline{\{y\}}$. Since X is T_0 , distinct points have different closures, so $x = y$. Therefore, $\leq$ is antisymmetric.

$(\Leftarrow)$ Suppose $\leq$ is a partial order. If $x \neq y$, then $x \not\leq y$ or $y \not\leq x$, so there exists an open set containing one but not the other. Hence X is T_0 . $\square$

5 Alexandroff Topology from a Preorder

Let $\leq$ be a preorder on X , and define:

$$\mathcal{B}(\leq) := \{\downarrow x := \{y \in X : y \leq x\}\}$$

Lemma 1. The collection $\mathcal{B}(\leq)$ is a basis for a topology $\mathcal{A}(\leq)$.

Proof. Every point $x \in X$ is contained in $\downarrow x \in \mathcal{B}(\leq)$, so the union of basis elements covers X . The intersection of two basis sets $\downarrow x \cap \downarrow y$ is $\downarrow z$ for some z , satisfying the basis criteria. $\square$

Lemma 2. A set $U \subseteq X$ is open in $\mathcal{A}(\leq)$ if and only if for all $x \in U$, $\downarrow x \subseteq U$.

Proof. If U is a union of $\downarrow x$, then for every $x \in U$, $\downarrow x \subseteq U$. Conversely, if $\downarrow x \subseteq U$ for all $x \in U$, then $U = \bigcup_{x \in U} \downarrow x$, which is a union of basis elements and hence open. $\square$

Theorem 2. $\mathcal{A}(\leq)$ is an Alexandroff topology.

Proof. Let $\{U_i\} \subseteq \mathcal{A}(\leq)$ be open. Let $x \in \bigcap U_i$. Then for all i , $\downarrow x \subseteq U_i$, so $\downarrow x \subseteq \bigcap U_i$. Hence, $\bigcap U_i$ is open. $\square$

6 Continuous Functions Between Alexandroff Spaces

Theorem 3. *Let $f : (X, \mathcal{A}) \rightarrow (Y, \mathcal{A}')$. The following are equivalent:*

1. f is continuous
2. If $x \leq y$ in X , then $f(x) \leq f(y)$ in Y

Proof. (1 $\Rightarrow$ 2): Suppose $x \leq y \Rightarrow x \in \overline{\{y\}}$. Then for any open set V containing $f(x)$, $f^{-1}(V)$ is open and contains x , thus intersects $\{y\}$, implying $f(y) \in V$. Hence $f(x) \in \overline{\{f(y)\}} \Rightarrow f(x) \leq f(y)$.

(2 $\Rightarrow$ 1): Suppose $x \leq y \Rightarrow f(x) \leq f(y)$. For any open set $V \subseteq Y$, $f^{-1}(V)$ is upward closed in the specialization preorder, hence open in X . $\square$

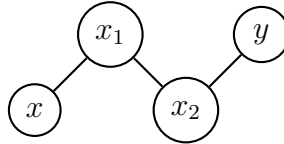
7 Connectedness of Alexandroff Spaces

Theorem 4. *If $(X, \mathcal{A})$ is Alexandroff, then it is locally connected.*

Proof. Since each point $x \in X$ has a minimal open neighborhood $V(x)$, and $V(x)$ is open and contains only points related via the preorder, the connected components are open. Therefore, the space is locally connected. $\square$

Theorem 5. *X is connected iff for all $x, y \in X$, there exists a sequence $x_0 = x, \dots, x_n = y$ with $x_i \leq x_{i+1}$ or $x_{i+1} \leq x_i$.*

Proof. This condition guarantees there is a path (in the preorder sense) from any point to any other. The existence of such paths ensures the space cannot be partitioned into disjoint nonempty open sets, hence it is connected. $\square$



Theorem 6. *An Alexandroff space is connected if and only if it is path-connected.*

Proof. In Alexandroff spaces, the preorder relation provides a natural notion of paths. If any two points can be connected via such a path, then the space is connected. Conversely, connectedness implies the ability to find such paths. $\square$

8 References

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